

Disciplina: Eletrônica Básica

Aula_1 - Diodos

Ambiente Blackboard

Apresentação

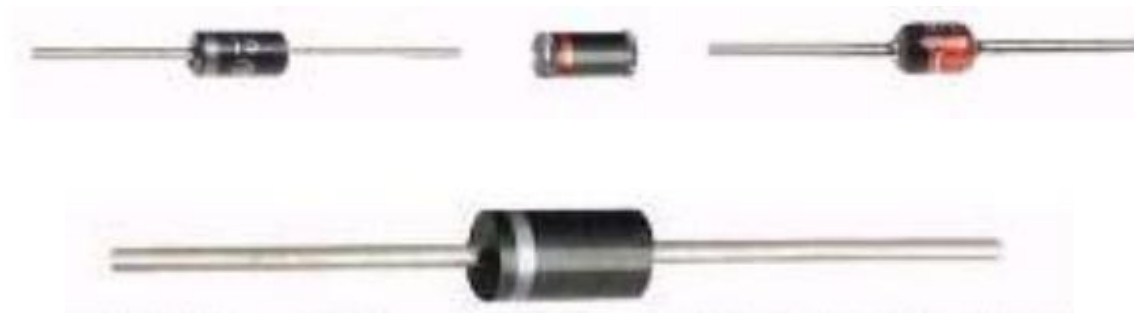
- 1) Diodo Ideal
- 2) Semicondutores
- 3) Diodo Real
- 4) Curva Característica
- 5) Tarefas – Exercícios
- 6) Modelos Equivalentes

Diodo

- Símbolo



- Exemplos

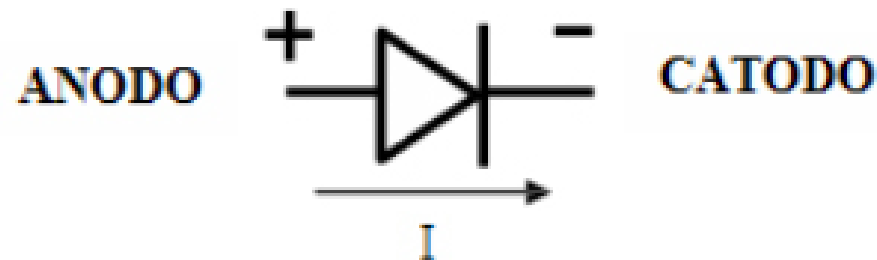


Diodo – Diodo Ideal

- Símbolo

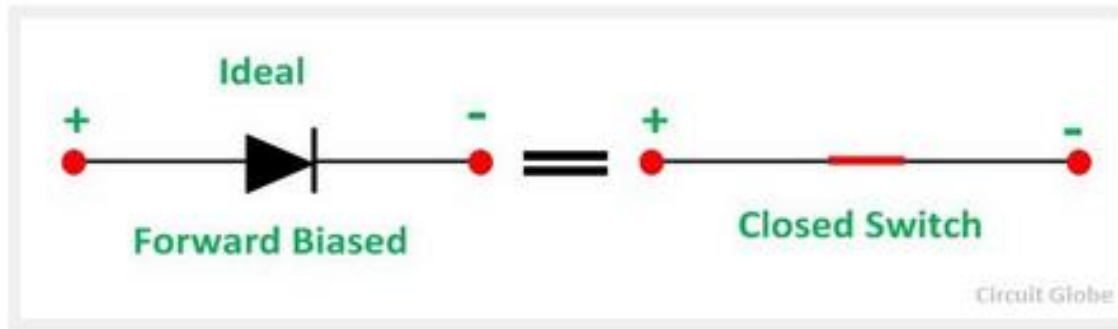


- Polarização

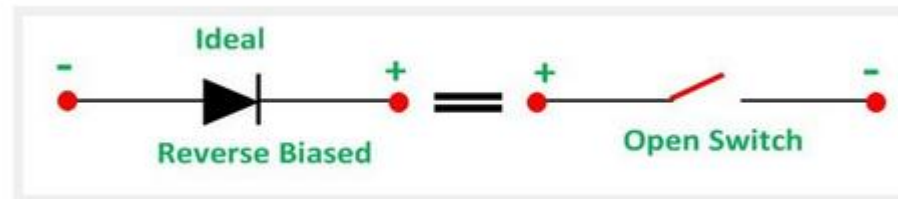


Diodo Ideal - Polarização

- Direta



- Reversa



Diodo Ideal - Resistência

$$R_D = \frac{V_D}{I_D}$$

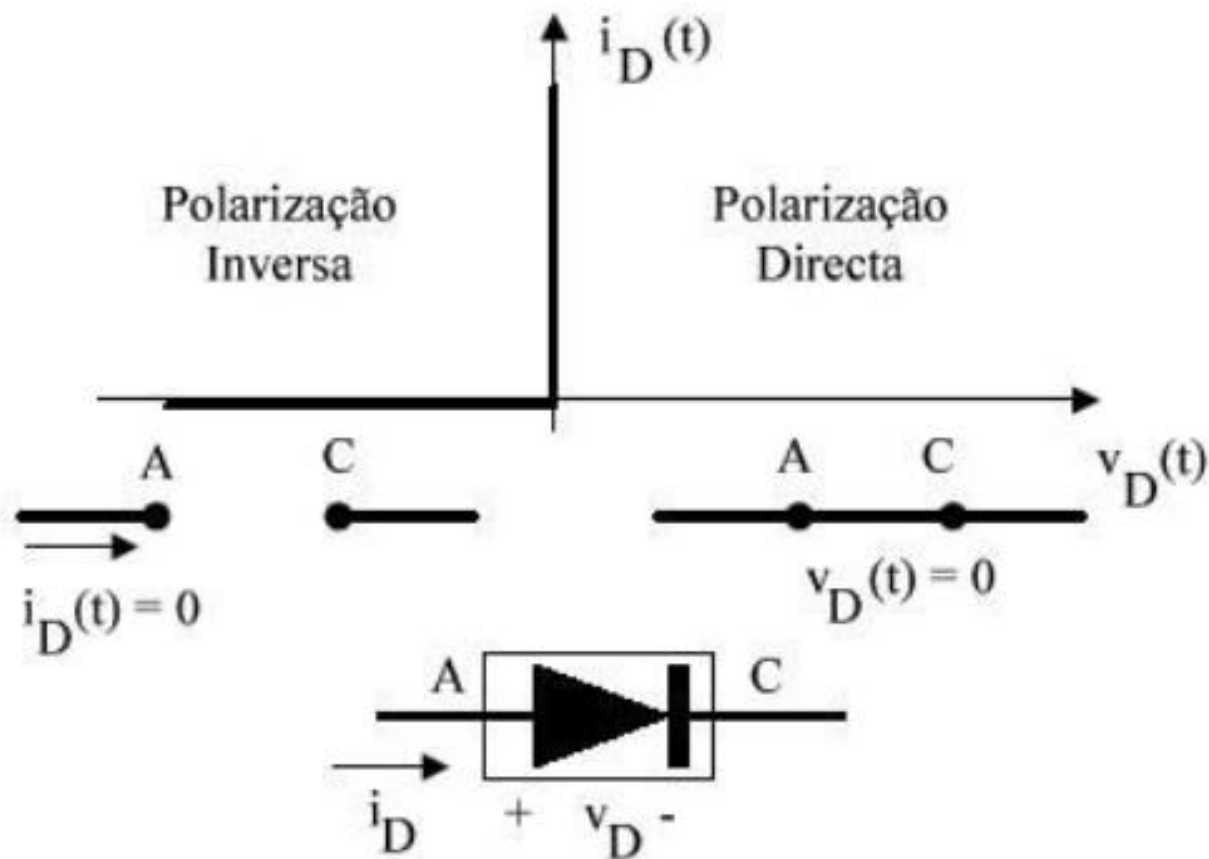
- Direta

$$R_D = \frac{V_D}{I_D} = \frac{0}{I_D} = 0\Omega$$

- Reversa

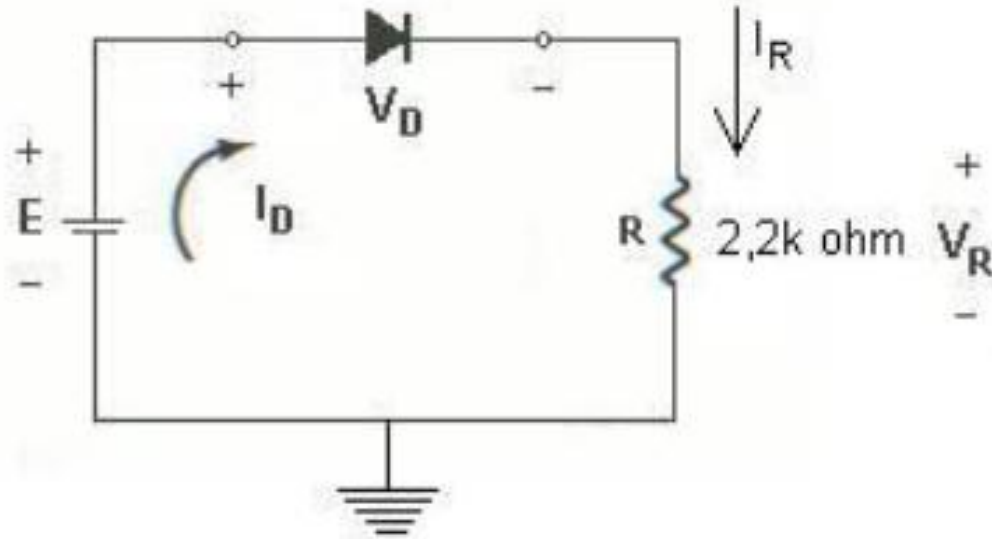
$$R_D = \frac{V_D}{I_D} = \frac{V_D}{0} = \infty$$

Diodo Ideal – Curva Característica



Diodo Ideal – Exercícios

Calcule I_D , I_R , V_D e V_R , para $E = 11V$. Considere o diodo ideal.

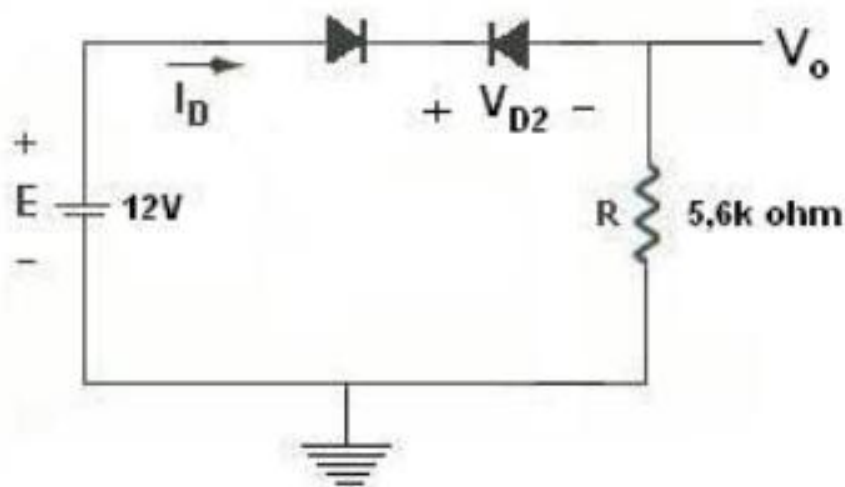


Diodo Ideal – Exercícios

Repita o exercício anterior considerando que a polaridade da fonte E foi invertida.

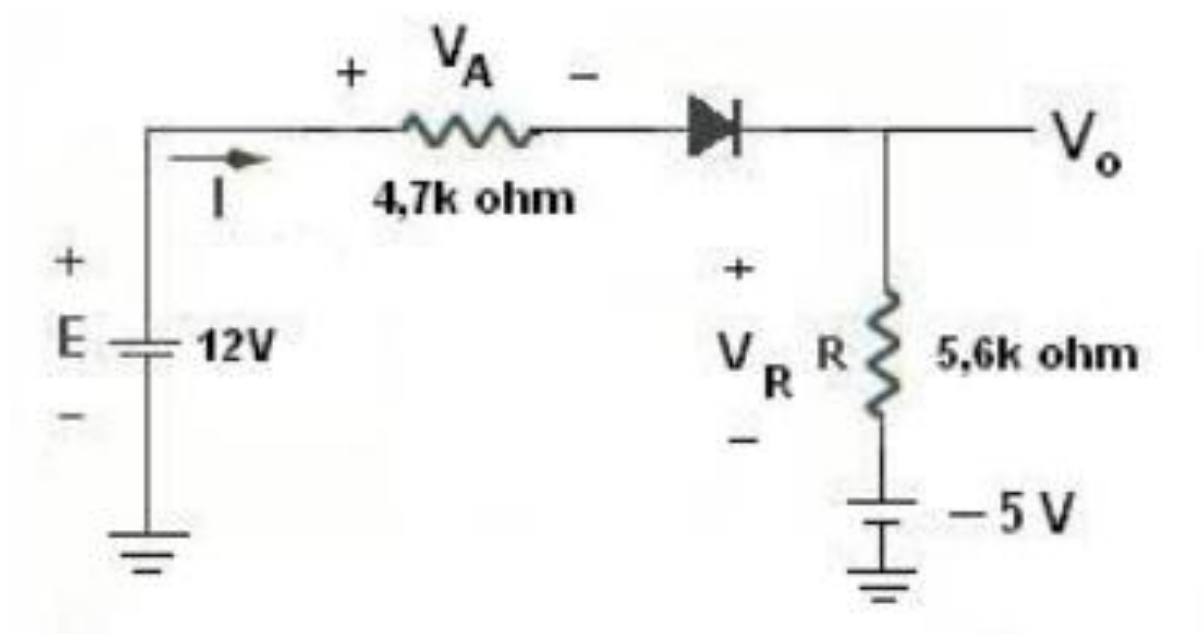
Diodo Ideal – Exercícios

Calcule I_D , V_o e V_{D2} . Considere diodo ideal.



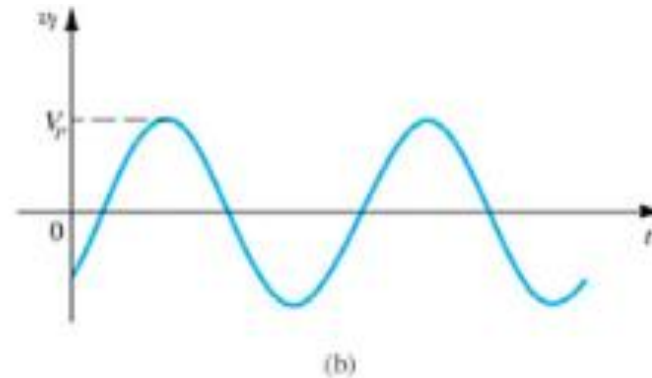
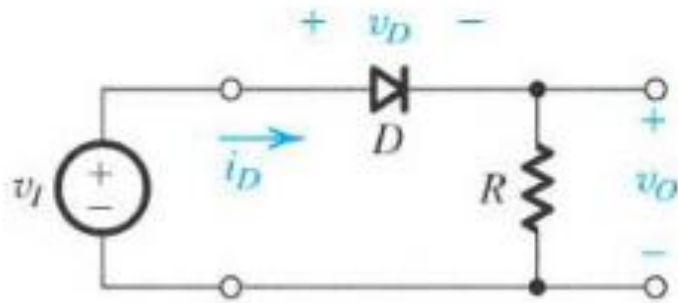
Diodo Ideal – Exercícios

Calcule I , V_A , V_R e V_o . Considere diodo ideal.



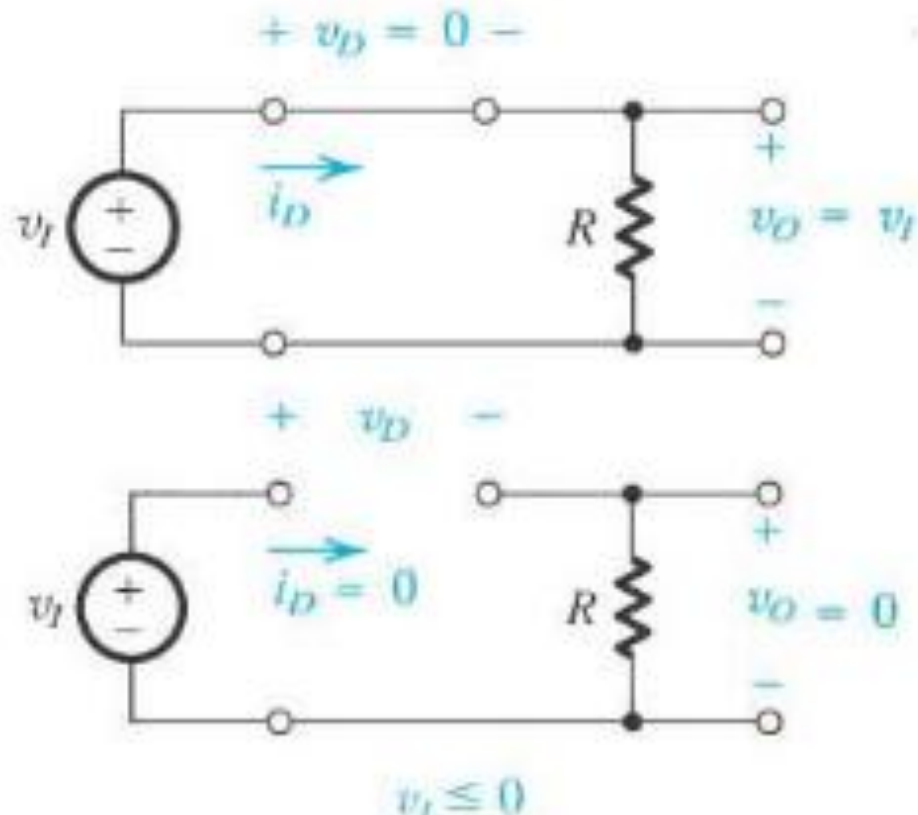
Diodo Ideal – Exercícios

Obtenha a forma de onda V_o para a entrada mostrada. Considere diodo ideal.



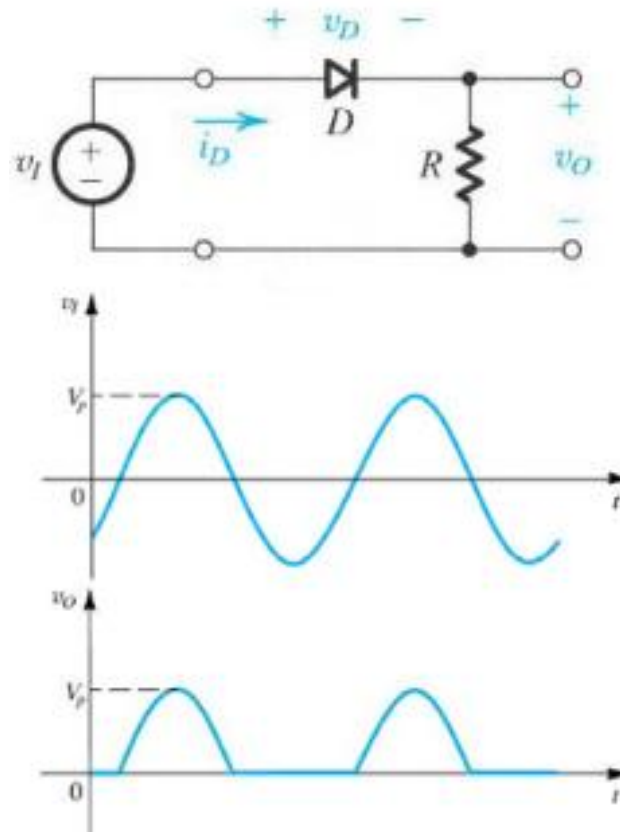
Diodo Ideal – Exercícios

Obtenha a forma de onda V_o para a entrada mostrada. Considere diodo ideal.



Diodo Ideal – Exercícios

Obtenha a forma de onda V_o para a entrada mostrada. Considere diodo ideal.



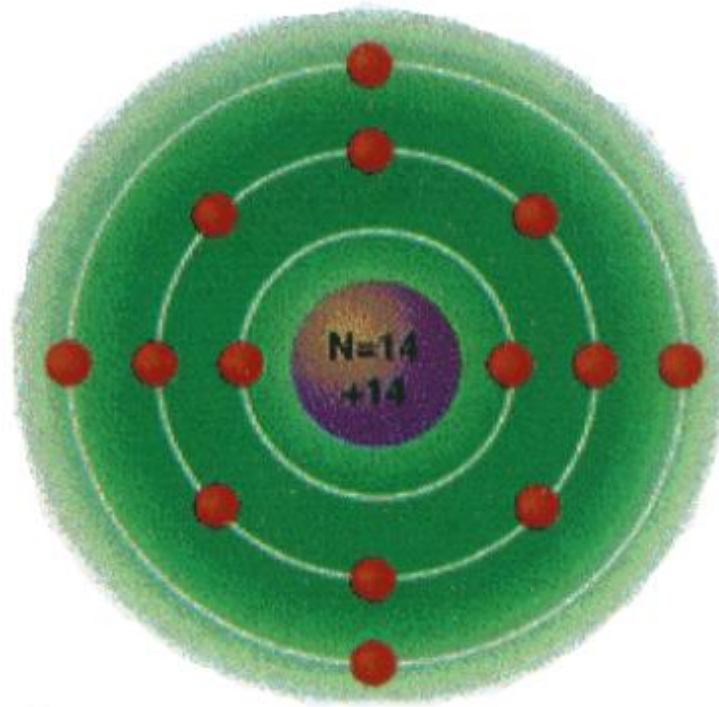
Semicondutores

- Silício (Si) e Germânio (Ge)
- Família 4A da tabela periódica
- 4 elétrons na camada de valência

5 B	6 C	7 N
13 Al	14 Si	15 P
31 Ga	32 Ge	33 As
49 In	50 Sn	51 Sb

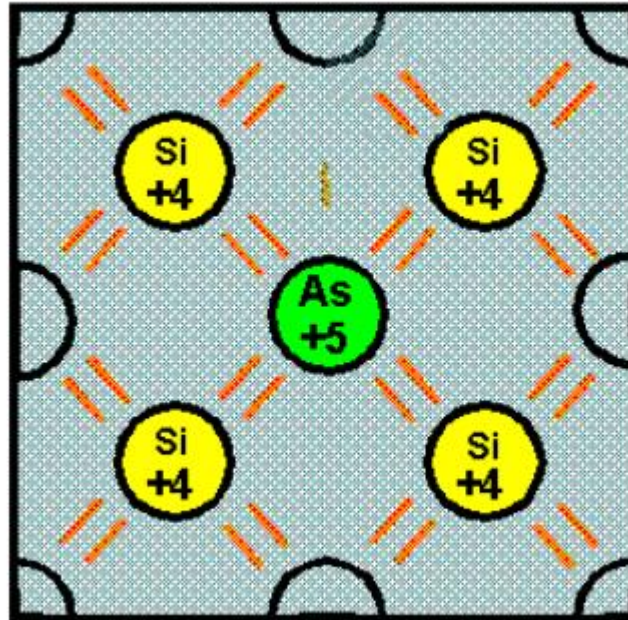
Semicondutores

- Silício (Si) – 4 elétrons na camada de valência



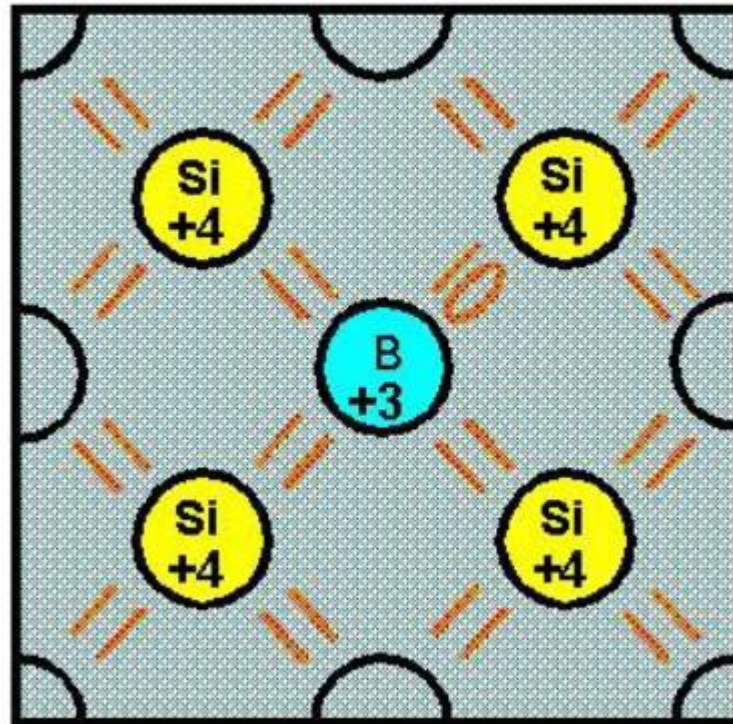
Semicondutores

- Dopagem – acréscimo de impurezas
- Tipo n



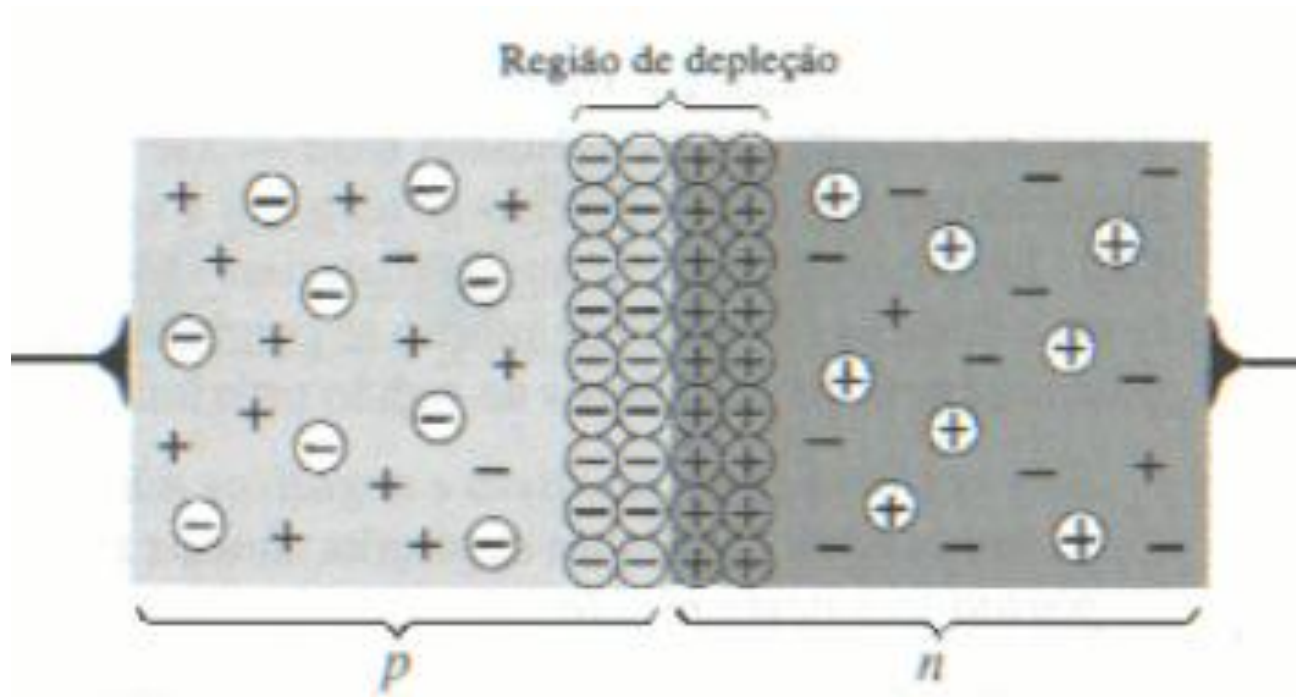
Semicondutores

- Dopagem – acréscimo de impurezas
- Tipo p



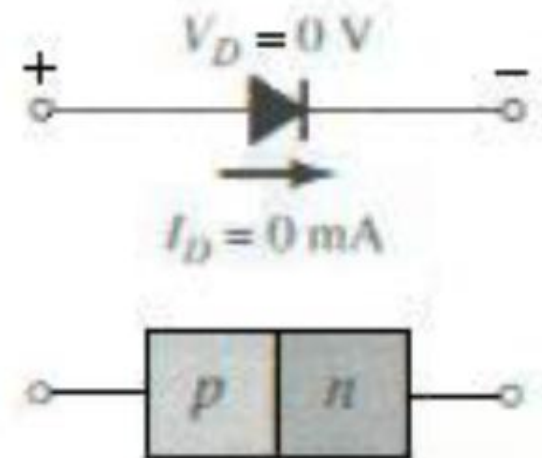
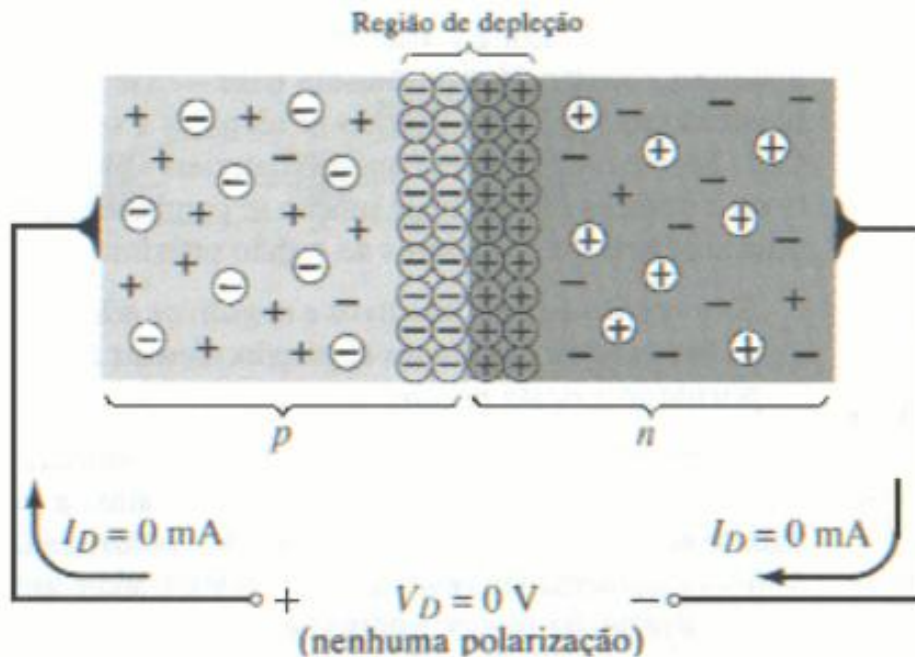
Semicondutores – junção PN

Diodo Real



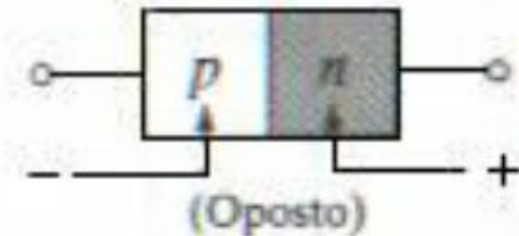
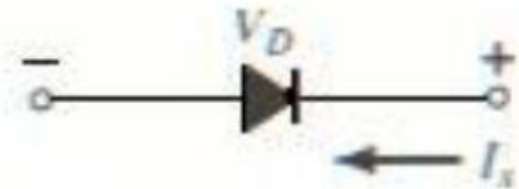
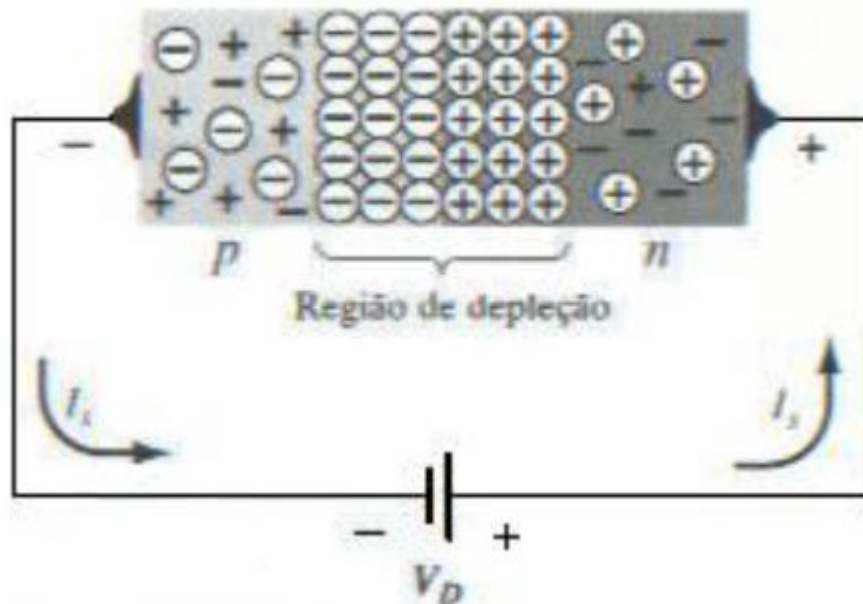
Diodo Real - Polarização

- Sem uma tensão aplicada.



Diodo Real - Polarização

- Aplicando uma tensão reversa.

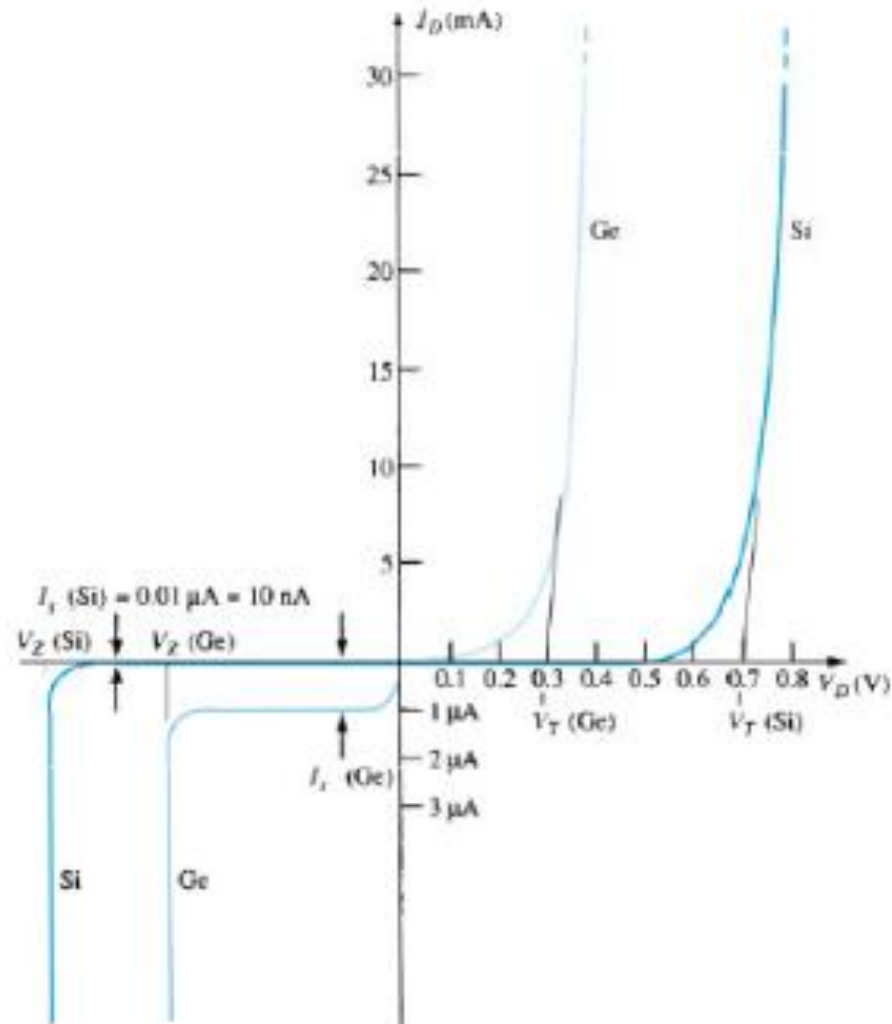


Diodo Real - Polarização

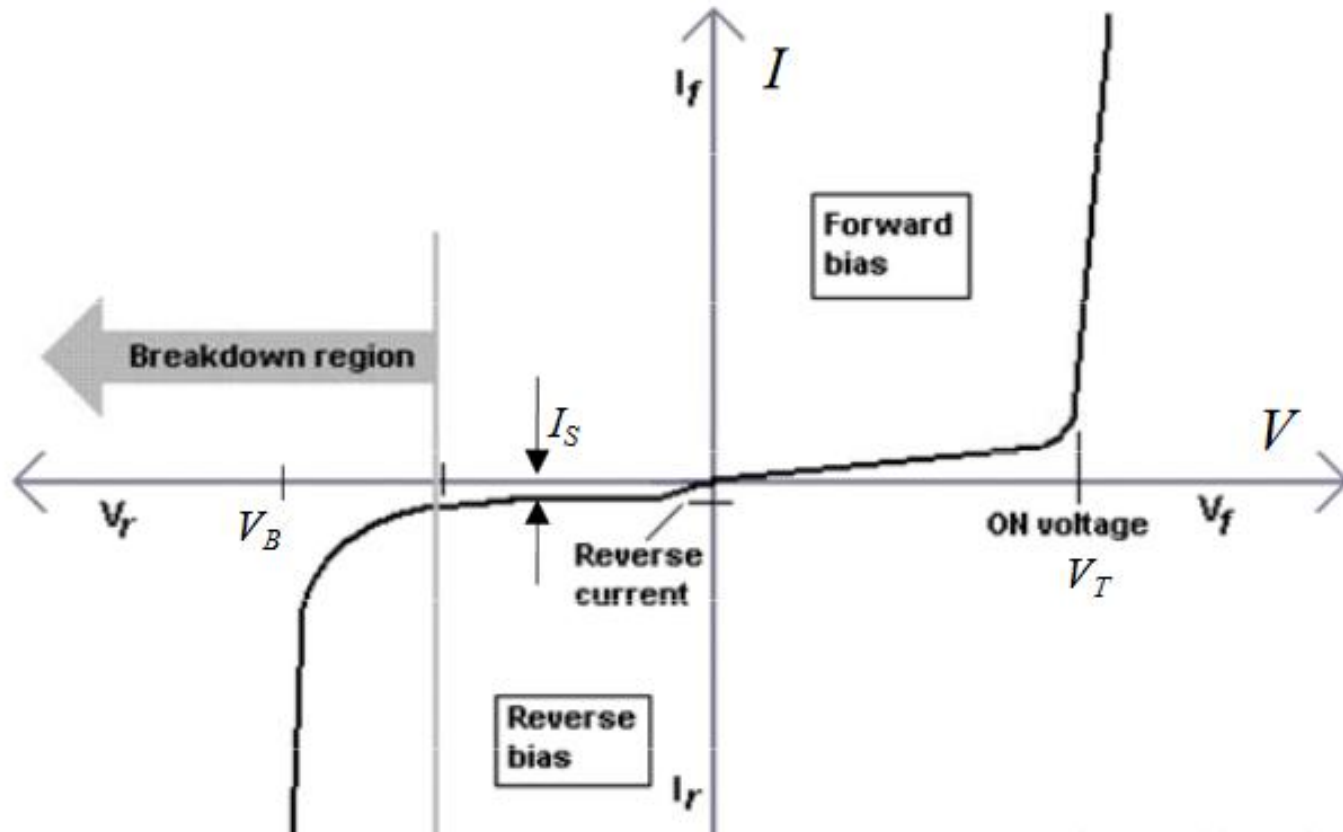
- Aplicando uma tensão direta.



Diodo Real – Curva Característica



Diodo Real – Curva Característica

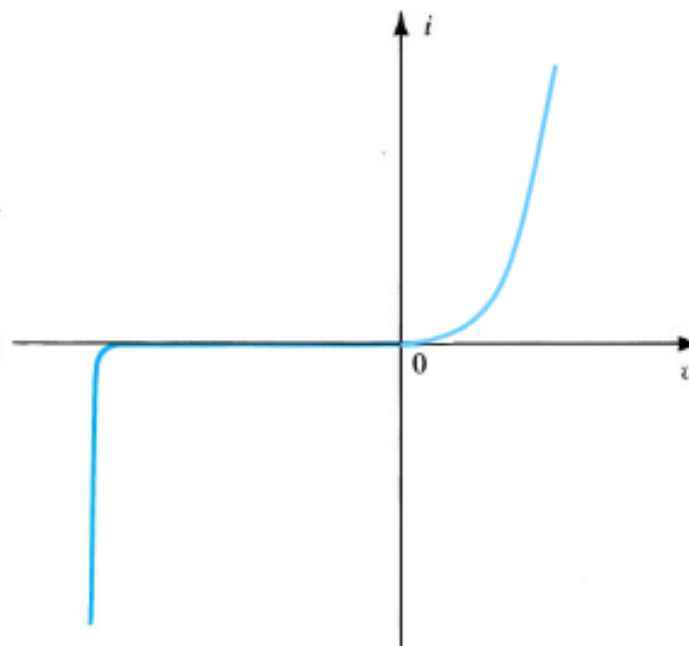


Diodo ideal X Diodo Real

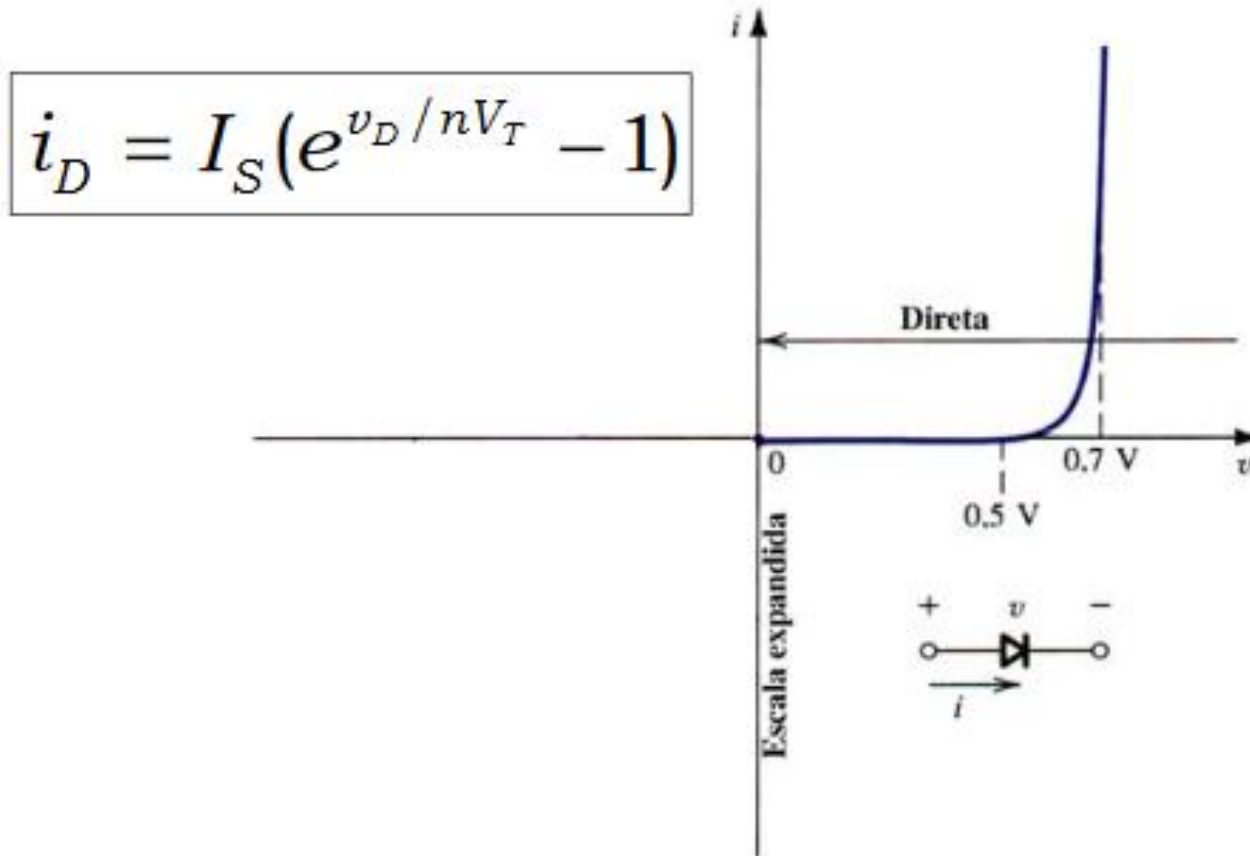
Diodo Ideal



Diodo Real



Diodo Real – Equação do diodo



Diodo Real – Equação da corrente

Equação da Corrente no Diodo (lei do diodo):

$$i_D = I_S (e^{v_D / nV_T} - 1)$$

Corrente no diodo Tensão no diodo

I_S = Corrente de saturação

n = fator de idealidade ($1 \leq n \leq 2$)

Tensão Térmica

$$V_T = k T / q$$

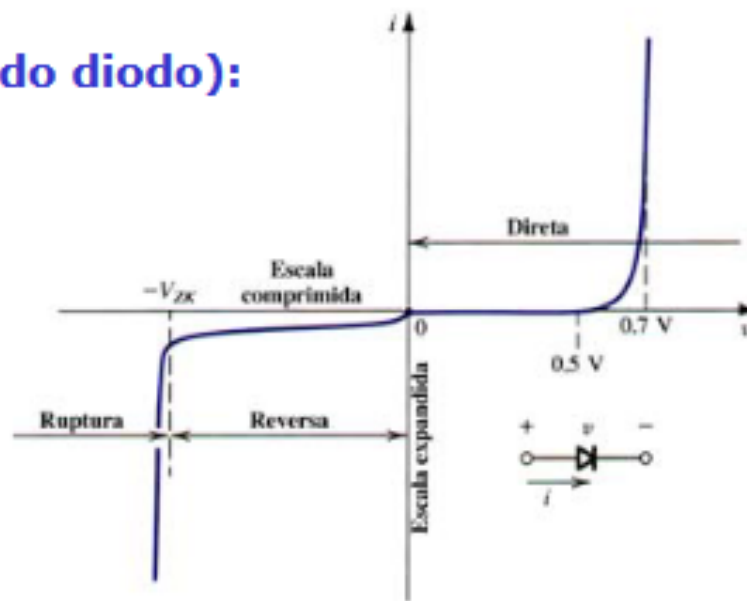
$$V_T = 25,8 \text{ mV} \quad (25^\circ\text{C})$$

$$V_T = \mathbf{25 \text{ mV}}$$

k = Constante de Boltzmann = $1,38 \times 10^{-23} \text{ J/K}$

T = Temperatura em kelvin = $(273 + T(^{\circ}\text{C}))$

q = carga do elétron = $1,6 \times 10^{-19} \text{ C}$



Folha de dados - Diodo Real

- <https://www.vishay.com/docs/88503/1n4001.pdf>



1N4001, 1N4002, 1N4003, 1N4004, 1N4005, 1N4006, 1N4007
www.vishay.com Vishay General Semiconductor

General Purpose Plastic Rectifier



DO-41 (DO-204AL)

FEATURES

- Low forward voltage drop
- Low leakage current
- High forward surge capability
- Solder dip 275 °C max. 10 s, per JESD 22-B106
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912



RoHS
COMPLIANT

PRIMARY CHARACTERISTICS

$I_{F(AV)}$	1.0 A
V_{RRM}	50 V, 100 V, 200 V, 400 V, 600 V, 800 V, 1000 V
I_{FSM} (8.3 ms sine-wave)	30 A
I_{FSM} (square wave $t_p = 1$ ms)	45 A
V_F	1.1 V
I_R	5.0 μ A
T_J max.	150 °C
Package	DO-41 (DO-204AL)

TYPICAL APPLICATIONS

For use in general purpose rectification of power supplies, inverters, converters, and freewheeling diodes application.

MECHANICAL DATA

Case: DO-41 (DO-204AL), molded epoxy body
Molding compound meets UL 94 V-0 flammability rating
Base P/N-E3 - RoHS-compliant, commercial grade

Terminals: matte tin plated leads, solderable per J-STD-002 and JESD 22-B102

E3 suffix meets JESD 201 class 1A whisker test

Tarefas

- Utilize o simulador de circuitos e obtenha a curva característica de diferentes diodos.
- Referências:
- [https://www2.seas.gwu.edu/~ece20/spring15/labs/tutorials/ECE 2115 Tutorial 1 DC Sweep Analysis in Multisim.pdf](https://www2.seas.gwu.edu/~ece20/spring15/labs/tutorials/ECE_2115_Tutorial_1_DC_Sweep_Analysis_in_Multisim.pdf)
- <https://www.youtube.com/watch?v=aGu8837gIKU>

Tarefas

- Resolva os exercícios apresentados nessa aula.
- Simule os circuitos utilizando um diodo ideal (virtual)
- Compare os resultados dos cálculos com os de simulação.

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